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Gazetteer- and type-enhanced Chinese address parsing BERT with an efficient global pointer

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ABSTRACT

As pivotal geospatial information, address data requires robust parsing to effectively support critical internet-based services, including navigation and e-commerce logistics. Yet, existing deep learning-based Chinese address parsing is still confronted by challenges arising from data annotation scarcity, administrative redistricting, regional variations, and informal expressions. Thus, this study enhances model generalization by leveraging contextual prior knowledge while fundamentally reforming the conventional paradigm that frames address parsing as a sequence labeling task. A Gazetteer- and Type-enhanced BERT with an efficient global pointer (GTBERT-Pointer) is proposed, representing an innovative architecture integrating a pre-trained BERT encoder, plug-and-play gazetteer adapter, type adapter, and enhanced global pointer mechanism. The BERT layer first derives the contextual vector representations of the address sequences. Then, the novel gazetteer and type adapters strategically infuse external gazetteer resources and hierarchical element-type knowledge into deep contextual embeddings. Finally, address parsing is recast as multi-label classification through the efficient global pointer to achieve global-span extraction of element boundaries. Empirical evaluations on recent Chinese address parsing benchmarks demonstrate the state-of-the-art performance of GTBERT-Pointer, surpassing existing approaches across comprehensive metrics.

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BERT; Chinese address parsing; global pointer; knowledge adapter

1 Introduction

With the rapid development of big data technology, smart cities, and logistics-navigation industries, address data has emerged as a core bridge between the physical space and the digital realm, empowering critical domains such as e-commerce logistics, emergency response, government registration, and financial transportation. However, such data often feature non-standardized expressions, dynamic evolution, and exponentially growing unstructured formats, imposing unprecedented challenges on automated textual address data processing. As the core task in Chinese address processing, address parsing is pivotal in enhancing urban digital management efficiency through improved accuracy (Lin, Kang, and He 2021).

Address parsing is a challenging task involving extracting structured information from unstructured data, i.e. mapping free-form textual addresses into discrete, structured semantic components and identifying their types (e.g. province, city, district, road, and house number) (Tian et al. 2016). Address parsing represents a particularly distinctive task in the context of East Asian languages, especially Chinese. Unlike Western language addresses that employ commas or line breaks for segmentation, Chinese addresses are complex expressions lacking explicit delimiters (Li et al. 2018). Moreover, these addresses often deviate from strict administrative hierarchies, demonstrating inconsistent regional patterns. Furthermore, unstructured Chinese address data in Internet mapping services may originate from diverse sources, thus exhibiting free-form textual expressions, abundant abbreviated aliases, and strong regional particularities (Zhang, Guonian, and Chen 2010). Due to these

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characteristics, Chinese address analysis constitutes a highly challenging yet research-worthy endeavor. As illustrated in Figure 1, the term '2小' ('two small') represents an abbreviated alias for '第二小学' ('the second primary school'), and a free-form expression in the second example is produced using road intersections for address description.

Early approaches to Chinese address parsing were rule-based (Guo et al. 2009; Zhang, Guonian, and Chen 2010) and statistical (Li et al. 2018; Ying et al. 2019), both struggling in complex real-world scenarios. Beyond static homogeneous addresses, numerous informal address expressions (e.g. abbreviations like '2小' for '第二小学', aliases, omissions, and free-form descriptions like road intersections) and regional variations influenced by local dialects, customs, and inconsistent hierarchical ordering can combine in potentially exponential ways. For instance, informal elements like slang or dialect-specific terms can interact with regional formats across China's diverse provinces and municipalities, resulting in a virtually unlimited number of combinations that defy exhaustive enumeration. This vast combinatorial space challenges the effectiveness of rule-based approaches that rely on predefined patterns and statistical methods that require comprehensive training data for modeling such variability. Since becoming the dominant paradigm in NLP (Liu et al. 2023; Jehangir, Radhakrishnan, and Agarwal 2023), deep learning models have progressively taken the place of conventional techniques of Chinese address parsing. Based on Conditional Random Fields (CRFs), Zhu et al. (2018) proposed a Chinese address resolution method that decomposes the address text into semantically independent elements, thus transforming parsing into a Named Entity Recognition (NER) task. To enhance performance, Zhang et al. (2020) introduced a BERT with a CRF architecture to identify novel address elements utilizing annotated address element data. As addresses are inherently hierarchical structured data, domain knowledge can be leveraged to improve deep learning methods. For instance, Zhang et al. (2023b) constructed sets of dynamic Finite State Machines (FSMs) to represent the multifaceted hierarchical structure of address data. Furthermore, Hu et al. (2024) employed geographic entity type knowledge to enhance the comprehension of intricate geospatial relationships inherent in textual descriptions.

While the above deep learning methods have significantly advanced Chinese address parsing, several critical challenges remain unresolved: First, the scarce and insufficient annotated domain-specific data hinders domain adaptation and results in ambiguities for distinguishing fine-grained semantic units, such as Building A and Building B in a park or the first and second entrances of a scenic area. Second, Chinese address parsing tasks involve few empty labels, unlike NER tasks, where empty labels typically outnumber entity labels. Address element diversity also means more out-of-vocabulary (OOV) words, complicating model validation. Third, Chinese address text elements often follow hierarchical orders from larger to smaller, a structural relationship that conventional NER methods cannot capture. Fourth, while deep learning models offer flexibility, their adaptability increases the risks of overfitting or hallucination when processing formal address expressions.

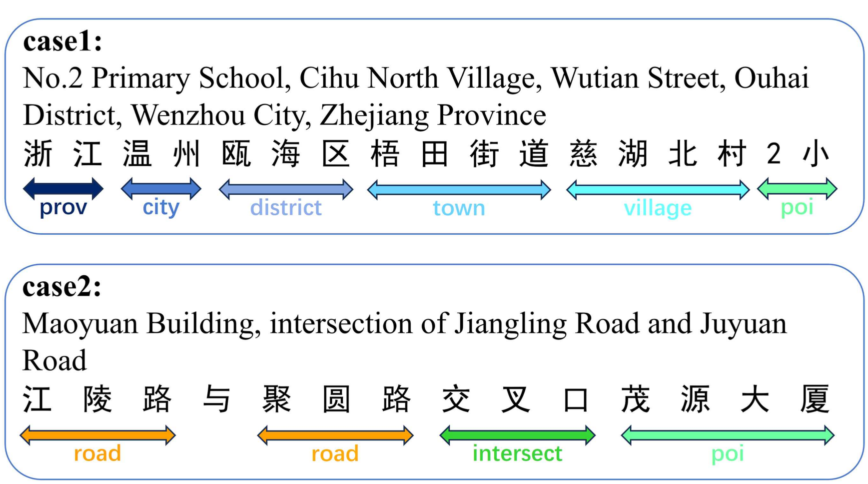


Figure 1. Examples of informal address expressions from CADD and GeoETA datasets.

Consequently, a deep learning-based parser that effectively integrates domain knowledge and rules is required for Chinese address parsing.

To address the aforementioned challenges, this study proposes an innovative Gazetteer- and Type-enhanced BERT with an efficient global pointer (GTBERT-Pointer) for Chinese address parsing. GTBERT-Pointer embodies three key innovations. First, the BERT encoder is pre-trained to process the address sequences as character-level inputs and generate contextualized dynamic character embeddings. Second, two novel BERT adapters are introduced. The gazetteer adapter employs a gazetteer-position attention mechanism to combine character features with spatial geographical information before leveraging bilinear attention for dynamic feature fusion. Integrating the Chinese gazetteer data solves the OOV issues due to imbalanced distributions and alleviates BERT's overfitting on generic address patterns. Meanwhile, the type adapter retrieves the five-level administrative divisions as a tree structure, adopting a character-type attention to fuse division words, hierarchical labels, and character representations. A dynamic updating mechanism is also adopted to prevent obsolescence from geopolitical redistricting. Then, the fused vectors are processed in subsequent BERT layers for computation, enabling parameter-efficient external knowledge learning. At the final stage, the efficient global pointer decoder transforms address parsing into a token-pair relation classification task. Compared to the original global pointer, it implements a shared scoring matrix for parameter efficiency and type-aware loss functions for high-cardinality labels. This adaptation better suits addresses with shorter sequences and numerous category labels. By systematically combining linguistic and geographical knowledge, balancing neural flexibility with rule-based constraints, and handling numerous entity labels, GTBERT-Pointer promises solid advantages. It achieves state-of-the-art performance on two open-source datasets, excelling in handling OOV issues, capturing hierarchical arrangement, and adapting to imbalanced address entity classes.

The primary contributions of this work are summarized as follows:

- The gazetteer and type adapters are devised for sophisticated, decoupled transformation of external address-related prior knowledge into vector representations, which are integrated with the contextual characterizations of address sequences to produce essential rule-based features for parsing.
- A modified efficient global pointer decoder is adopted, which achieves superior adaptability to the unique requirements of address parsing.
- GEBERT-Pointer is enhanced with a dynamic weighting method to fuse character-type and character-gazetteer representations and a shared scoring matrix with type-aware loss functions in the efficient global pointer, thus improving parameter efficiency while addressing the high-cardinality labels in multi-category address parsing.

2 Related work

This work closely relates to the existing address parsing neural networks, knowledge-based methods, and Pointer Networks, a brief overview of which is provided below.

2.1 Neural networks for Chinese address parsing

Aiming at complex Chinese address parsing, Lu, Liu, and Zhou (2019) designed a neural network with Gated Recurrent Unit (GRU) models under a sequence-to-sequence (seq2seq) framework. Their method directly standardized irregular address inputs via end-to-end prediction but lacks interpretability compared to other approaches. To overcome the limitation of purely linear sequential structures in capturing dependencies between non-adjacent address element labels, Liu et al. (2019) proposed a chart-based neural network parser that incorporates latent variable tree structures. Their parser captures the head-dependent relationships based on the original linear chain structure of the addresses while modeling tail dependencies via latent variable trees, performing comparably to Long Short-Term Memory (LSTM) models.

Pre-trained language models (PLMs) have outperformed traditional tagging-based methods in NER tasks (Devlin et al. 2019; Liu et al. 2019). Thus, Zhang et al. (2020) applied the BERT model to Chinese address parsing, leveraging its bidirectional encoder representations to capture contextual information and Chinese address features while employing CRFs to model constraints between labels. Subsequently, Zhang, Luo, and Wu (2023) replaced BERT with the RoBERTa model, which is better suited for the Chinese context. By inserting a BiLSTM layer between the RoBERTa encoder and CRF decoder to filter hidden state information while retaining features useful for downstream decoding, their RoBERTa-BiLSTM-CRF architecture achieved superior address parsing to BERT-CRF.

As another branch of PLMs, Large Language Models (LLMs) like GPT−4 (OpenAI et al. 2024) and LLaMA−3 (Grattafiori et al. 2024) excel in multi-task scenarios via scaling enabled by their decoder-only Transformer architecture, which allows almost unlimited contextual content modeling. Yet, evaluation of the zero-shot performance of Llama2-7B and Mistral-7B on address parsing data revealed non-negligible hallucinations in LLMs, including diverting from instructions, incorrect predictions, and generating labels not included in the input (Hammami, Baligand, and Petrovski 2024). Based on QWen-7B, AddrLLM (Yang et al. 2025) integrates instruction fine-tuning to align with downstream tasks, including address parsing, Address Entity Prediction (AEP), and address rewriting. However, it incurs prohibitively high full fine-tuning costs. CapICL (Ling et al. 2023) utilizes BERT-based similarity computation to retrieve relevant samples and employs GPT−3 for address parsing, using prompts constructed by combining examples, original text, and explanatory text. Nonetheless, CapICL depends constantly on BERT weights, and its prompt-engineered LLM underperforms even BERT-based models in address parsing. In summary, LLMs enable multi-task learning and complex reasoning in NLP with limited samples but are ill-suited for solving singular tasks like address parsing.

While these methods achieve word segmentation and NER primarily via NLP techniques, Geo-BERT (Liu et al. 2021b) realizes geographical awareness with a novel BERT pre-trained using massive Point-of-Interest (POI) texts and graph representations, thereby extracting the corresponding cities, districts, towns, or roads from POI addresses. MGeo (Ding et al. 2023) further incorporates real-world geographical locations as an additional modality and jointly maps textual and spatial information into dense representations during pre-training. The geospatial features integrated enhance the understanding of the administrative hierarchy, fine-tuning the model for the Chinese address segmentation downstream. However, its training and fine-tuning costs are prohibitively high for solely Chinese address parsing, thus hindering specialized implementations.

2.2 Knowledge based methods

The full-parameter fine-tuning of large pre-trained models risks suboptimal performance and substantial resource consumption. The knowledge-enhanced approach provides pre-organized domain knowledge to mitigate the reliance of PLMs on retraining. Zhang et al. (2023a) proposed a graph-based address representation framework for address parsing, which transforms address information into graph models, and validated it for address matching and toponym disambiguation within graph databases. By exploiting the hierarchical structure of addresses, the dynamic functional state machine method (Zhang et al. 2023b) constructs collections of dynamic FSMs as a knowledge repository for sequencing and composing elements in Chinese addresses, thus improving address parsing. Leveraging existing relation extraction architectures, Hu et al., 2024 introduced the GeoWISE framework to extract geospatial topological relations, which effectively integrates semantic comprehension and external geospatial domain knowledge. GeoWISE enhances pertinent tasks by aligning external knowledge with contextual information via Natural Language Inference (NLI), thereby leveraging in-depth geospatial expertise.

Domain-specific adaptation requires substantial modifications to a considerable portion of model parameters, often costing the pre-trained PLMs the capabilities for other tasks. Accordingly, Houlsby et al. (2019) adapted a pre-trained model to NLP tasks by modifying or adding a small subset of its parameters, a technique known as parameter-efficient fine-tuning. Adapters in transformers first project the layers to a lower-dimensional space, apply a nonlinear activation, and then project back to the original dimension. These BERT adapters enable task-specific adaptation without base model parameter modifications, thus preserving architectural integrity while avoiding massive parameter expansion. K-ADAPTER (Wang et al. 2021) pioneered knowledge injection through adapters and performed well in knowledge-intensive tasks like relation classification. This success translates to lexicon integration, as demonstrated by Chinese NER through lexicon infusion.

SoftLexicon (Ma et al. 2020) is another decoupled approach that preserves pre-trained models while enhancing character representations with lexicon features. However, its higher character embedding dimensions arising from the 5-dimensional multi-hot vector representation for lexicon knowledge substantially elevate training complexity. ZEN (Diao et al. 2020) addresses this limitation by integrating n-gram representations via BERT adapters, whose fixed-length modeling, however, hinders their lexical information capacity. LEBERT (Liu et al. 2021a) introduces lexicon knowledge through K-Adapter-like decoupled modules but is prone to lexical enhancement uncertainty and lacks positional information to associate enhanced sequences. TypeBERT (Li et al. 2023b) pre-injects lexical type information but requires dataset-specific configurations. While effective, the required dataset-specific category configurations increase training costs.

Eventually, direct lexicon injection proves inadequate for parsing Chinese addresses. Inspired by LEBERT and TypeBERT, we propose injecting gazetteer and administrative division knowledge into BERT models. Our innovative integration of a spatial-aware attention mechanism within the gazetteer adapter provides essential positional context for geographic entities.

2.3 Pointer networks

Pointer networks, a variant of the Seq2Seq learning framework with the encoder and decoder RNNs enhanced by attention mechanisms, suit particularly well for tasks requiring source-to-target sequence element copying (e.g. text summarization (See, Liu, and Manning 2017) and machine translation (Vaswani et al. 2017). Leveraging this mechanism, (Zeng et al. 2018) proposed a Seq2Seq model integrating copy mechanisms and graph convolutional networks, successfully adapting pointer networks to NER tasks through joint extraction. However, these approaches cannot effectively capture global semantic features during entity localization. To address this limitation, Su et al. (2022) enhanced traditional pointer networks with a global pointer network that treats entity boundaries as integrated units and incorporates rotary position embedding and a relatively unbiased multi-label classification loss function. The global pointer network enables global perspective prediction of entity spans while maintaining superior performance over CRF-based methods. Subsequent improvements include enhanced robustness via adversarial training (Li et al. 2023a) and the RoBERTa-Global Pointer Zhang, Luo, and Wu (2023) that combines character-level (RoBERTa) and word-level representations for Chinese judicial NER.

Despite these advancements, directly applying a global pointer to parsing Chinese addresses with numerous categories introduces excessive parameters and training complexity. To overcome these limitations, the efficient global pointer implements a shared scoring matrix across entity categories to improve parameter efficiency and incorporates the pre-segmentation results into the loss function to better accommodate the high-category scenarios. These innovations maintain the global prediction advantage while improving suitability for parsing Chinese addresses with more entities and categories.

3 Methodology

The backbone structure of the proposed GTBERT-Pointer is illustrated in Figure 2. This section elaborates on GTBERT-Pointer from five perspectives: gazetteer embedding corpus construction, gazetteer adapter, type adapter, GTBERT, and efficient global pointer. The gazetteer embedding corpus is constructed based on Chinese toponym articles. The gazetteer adapter derives the gazetteer vectors from the input address character sequences and the gazetteer embedding corpus. The derived gazetteer vectors are then fused with the character sequence representations produced by the first-layer BERT to generate gazetteer and character hybrid vectors. Similarly, the type adapter computes the type and character hybrid vectors for the input address character sequence. GTBERT computes address sequence representations at its initial layer and inputs them simultaneously into the gazetteer and type adapters to generate two hybrid representations. Subsequently, a dynamic weighting method is employed to produce the final unified

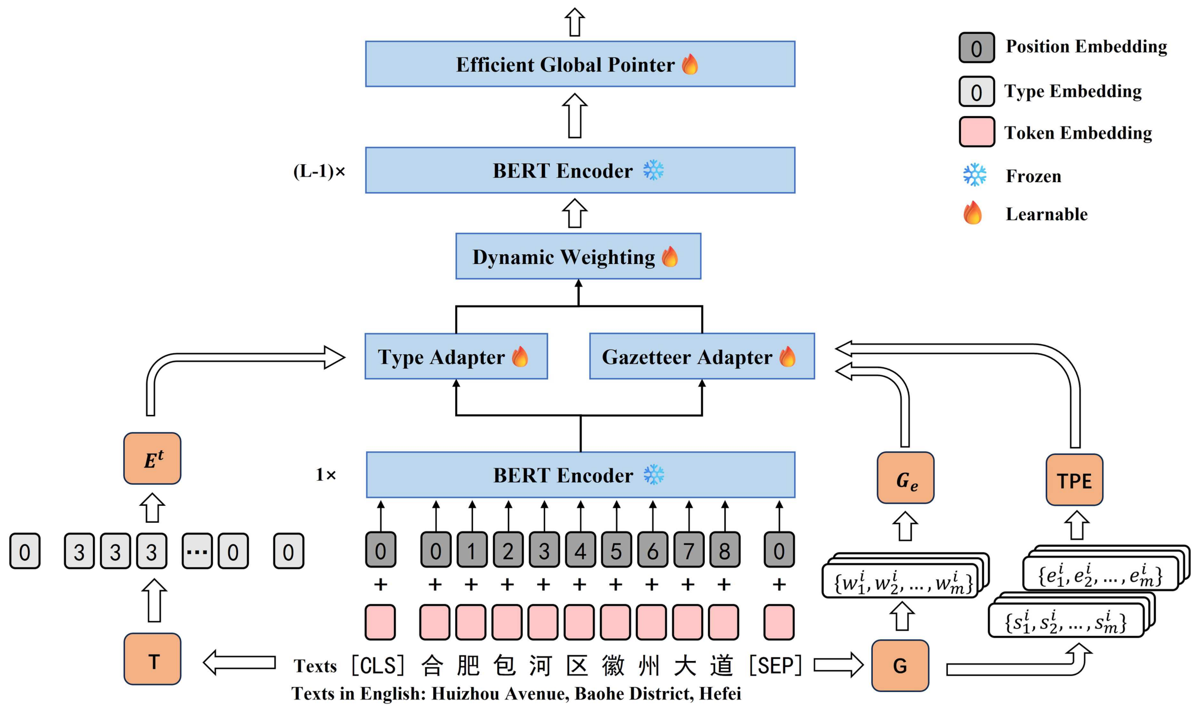


Figure 2. The backbone architecture of GTBERT-Pointer, where L means the number of BERT layers, G and T represent the gazetteer static embedding corpus and administrative division data, G_e and E^t are gazetteer embedding table and type embedding, TPE is short for trigonometric positional encoding, and the detailed structures of the gazetteer adapter, type adapter, and efficient global pointer are shown in Figure 3.

representation for calculations in subsequent layers. Finally, the efficient global pointer determines the start and end positions of the address entities from a global perspective alongside label classification, based on the GTBERT output.

3.1 Gazetteer adapter

Character-based BERT encoders have seen wide applications in Chinese NLP tasks, which treat individual characters as basic units. However, their lack of lexical semantics often causes poor performance in specialized domains. Thus, the gazetteer adapter embeds candidate toponyms into the character-level representations of standard BERT encoders to facilitate the learning of word-level features and incorporate domain-specific knowledge for tasks involving addresses. It also encodes positional information for toponym elements to improve the detection of spatial relationships, entity boundaries, and feature lengths, thereby aiding element importance assessment. Inspired by the adapter architecture of LEBERT, the gazetteer adapter is inserted behind the first BERT layer, allowing it to fuse external knowledge vectors with hidden character representations to produce augmented states for subsequent layers. Thus, efficient knowledge integration is realized with minimal parameter changes.

A gazetteer static embedding corpus G is constructed with geographic names and contextual descriptions from georeferenced sources to provide pre-computed word embeddings for integrating geographic knowledge into the gazetteer adapter. The source for corpus G is Wikipedia articles, with each entry formatted as ‘geographic name + summary’ following segmentation using Jieba to enhance domain-specific representations. Embeddings are generated using FastText’s Skip-gram strategy, which decomposes words into subwords (for instance, '北京市' into '北', '京', and '北京市') to facilitate the processing of rare toponyms and neologisms. Data collection and cleansing are detailed in Appendix A. The corpus G constructed contains 533,824 entries, comprehensively covering toponymic elements in Chinese address parsing datasets. The embedding quality is evaluated through intra-class and inter-class similarity metrics. The intra-class similarity measures vector distances among toponyms within the same administrative

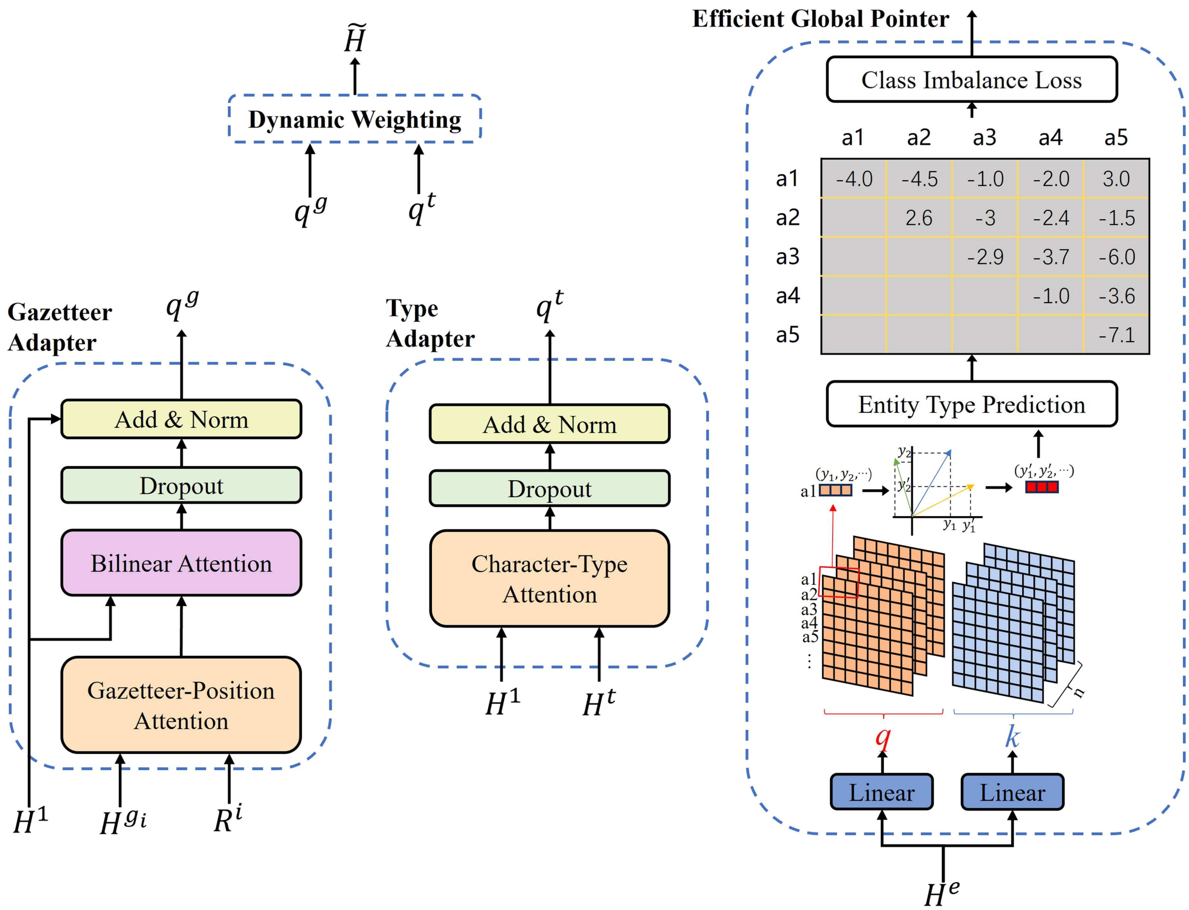


Figure 3. A detailed schematic diagram of the gazetteer adapter, type adapter, dynamic weighting module, and efficient global pointer, all variables in which are elucidated in Section 3.

hierarchy (e.g. province-to-province), ensuring high clustering cohesion. The inter-class similarity maintains sufficient distinction across administrative levels. These evaluations demonstrate the model's generalization capability for common, obscure, and OOV toponyms. Additionally, non-geographic features are incorporated from external lexicons such as the Tencent AI Lab Static Embedding Corpora (Song et al. 2018).

With the gazetteer embedding corpus established, the gazetteer-position pair sequence is generated based on the Chinese address sequence a and gazetteer corpus G, forming the standard input for the gazetteer adapter. By preserving the matching relationship between the gazetteer word embeddings and their corresponding positions in the original sequence, this sequence enables the adapter to integrate additional gazetteer features into the address sequence. The relative positions of candidate gazetteer terms in the address sequence and their lengths significantly influence the integration effectiveness of external gazetteer features during fusion. For address sequence $a = \{a_1, a_2, \ldots, a_n\}$, m potential candidate gazetteer terms $w^i = \{w_1^i, w_2^i, \ldots, w_m^i\}$ are retrieved from G for each character a_i. while maintaining start and end position sequences s^i and e^i of equal length to record candidate term placements, these positions are concatenated with the candidate terms to form the gazetteer-position pair sequence for a_i: $wse^i = \{(w_1^i, s_1^i, e_1^i), (w_2^i, s_2^i, e_2^i), \ldots, (w_m^i, s_m^i, e_m^i)\}$. Each character is represented by a two-dimensional list of all potential words and their corresponding head and tail indices. Explicitly incorporating positional information enhances the model's capability to determine whether these terms constitute essential address components.

The gazetteer-position attention module is the first module of the gazetteer adapter. It takes a standard gazetteer-position pair sequence as input and integrates the word vectors of candidate toponyms with their spatial positional information before fusing sequence characters and candidate toponym features. Its output is a hybrid candidate toponym word vector containing spatial positional information to help the model capture the order and significance of lexical information. For specific implementation, the module first employs a static embedding table G_e to produce word embeddings for each candidate toponym w_j^i in the candidate toponym sequence w^i from the gazetteer-position pair sequence:

$$h_j^{g_i} = G_e(w_j^i), \tag{1}$$

where G_e is the word lookup table, and $h_j^{g_i}$ represents the embedding vector of the j-th candidate geographical name for the i-th character in the sequence. All $h_j^{g_i}$ collectively constitute the geographical name embedding list $H^{g_i} = \{h_1^{g_i}, h_2^{g_i}, \ldots, h_m^{g_i}\}$.

Similarly, the positional sequences s^i and e^i in the gazetteer-position pair sequence are converted into embedding vectors using an explicit encoding method for transformer models, i.e. trigonometric positional encoding, to derive the start and end positional encodings. The trigonometric positional encoding is calculated in Equation (2):

$$p_{i,2k+1} = \sin\left(\frac{i}{10000^{2k+1/d}}\right), \tag{2a}$$

$$p_{i,2k} = \cos\left(\frac{i}{10000^{2k/d}}\right). \tag{2b}$$

Following trigonometric positional encoding, the corresponding elements of the first and last positional vectors are summed to represent the relative positional embedding vector list $R^i = \{R_1^i, R_2^i, \ldots, R_m^i\}$ for each character:

$$R^i = p_{s_i} + p_{e_i}, \tag{3}$$

where + denotes vector addition, and p_{s_i} and p_{e_i} correspond to the list of start and end vectors for candidate words at the i-th character in the address sequence.

The two encoding processes above produce the two inputs of the gazetteer-position attention module: the word embedding vector list h^i and the position embedding vector list R^i. The gazetteer-position attention mechanism calculates the linear attention between gazetteer word embeddings and relative position vectors to derive the weighted hybrid hidden state representations. As positional information cannot be captured by directly applying self-attention mechanisms to candidate word vectors, resulting in an input sequence order not influencing the judgment of candidate term importance, we propose the position-aware attention. This mechanism allows dynamically weighing the importance of different candidate toponyms based on their positions within the sequence. The gazetteer-position attention mechanism can be expressed as Equation (4):

$$x_j^i = W_1(ReLU(W_2 h_j^{g_i} + W_R R_j^i + b_2)) + b_1, \tag{4}$$

where W_1 and W_2 are learnable weight matrices, b_1 and b_2 are the corresponding fixed biases, subscript j indicates the j-th element of the corresponding vector list, and ReLU denotes the rectified linear Unit function.

The bilinear attention module with a specialized additive attention mechanism is the second module of the gazetteer adapter. By fusing the hidden state vectors of sequential characters with those of the corresponding candidate toponyms, the bilinear attention module produces enhanced hidden state vectors for the sequence characters. Such enhanced vectors are reintroduced into the BERT encoder for the remaining training process. The primary function of the bilinear attention module is to evaluate the importance of different candidate word vector representations relative to the corresponding characters, thereby facilitating the determination of address element boundaries and categories during word segmentation. The gazetteer adapter is inserted behind the first BERT encoder layer and takes the hidden state vectors it generated for character sequence a as input. The character sequence a initially undergoes character embedding processing, followed by transformation through the first BERT encoder layer to derive the hidden state vectors $H^1 = \{h_1, h_2, \ldots, h_n\}$, as formulated in Equation (5):

$$\boldsymbol{H}^1 = f^1(E^c(\boldsymbol{a})). \tag{5}$$

The input to the bilinear attention module consists of the hidden state vector and its corresponding hybrid hidden state vector of candidate gazetteer entries. Each bilinear attention computes a character-gazetteer hybrid vector for each character in the sequence, ultimately forming a character-gazetteer hybrid vector list with a length of n. The calculation formulas for the bilinear attention module are written in Equation (6):

$$s_i^g = h_i W_g x_j^{iT}, \tag{6a}$$

$$a_{ij}^g = \exp\left(s_{ij}^g\right) / \Sigma_{j=1}^m \exp\left(s_{ij}^g\right), \tag{6b}$$

$$q_i^g = \Sigma_{i=1}^n a_{ij}^g x_j^i, \tag{6c}$$

$$q^g = \Sigma_{i=1}^n q_i^g \times h^{g_i}. \tag{6d}$$

In Equation (6a), W_g is a learnable weight matrix that learns the importance of different candidate gazetteer vectors, where h_i and x_j^i represent the i-th character vector and the j-th candidate gazetteer vector corresponding to the i-th character, respectively. s_i^g and a_{ij}^g denote the corresponding candidate gazetteer scores and the normalized scores, respectively. Finally, the weighted character-gazetteer representation q^g is generated through score-weighted aggregation.

3.2 Type adapter

The type adapter is designed to enhance the parsing of formal addresses by integrating hierarchical administrative division categories with character information from address sequences. It processes simplified hierarchical category vectors and functions in a similar manner to a streamlined gazetteer adapter. Employing a character-type attention mechanism, it fuses administrative division hierarchy vectors with address character sequences. Prior to fusion, hierarchical sequences are constructed for each address based on the Statistical Classification of Administrative Regions and Urban-Rural Divisions (2023) from the National Bureau of Statistics of China (NBSC)[1]. This document categorizes administrative units into five levels: provincial (provinces, municipalities, autonomous regions), prefectural (cities), county (districts and counties), township (townships and subdistricts), and village (villagers' committees and residents' committees). To accommodate informal address variations, the data are augmented using regular expressions to map official names to common forms and abbreviations, e.g. `山东省' (Shandong Province), `山东' (Shandong), and `鲁' (Lu).

Based on the Chinese address sequence a and administrative division data T, a type sequence is derived, which encapsulates hierarchical administrative information. Administrative division levels are assigned integers 1 to 5, with larger numbers indicating finer divisions. Meanwhile, non-administrative components are marked as 0. The type sequence is constructed with exact substring matching enabled by regular expressions. Starting from the beginning of the address, only the longest match for each administrative level is retained to ensure each level appears consecutively only once while prohibiting finer-level elements preceding coarser ones. This process generates a type sequence t with the same length as a.

The type adapter also integrates a six-dimensional type embedding mapping alongside conventional character-based BERT components (token embeddings, segment embeddings, and position embeddings). This novel embedding layer projects administrative division levels into the same dimensional space as the character embeddings through linear transformation, thus deeply fusing administrative hierarchy vectors and character representations. This operation integrates the hierarchical administrative information into the embedding system while maintaining dimensional consistency.

$$\boldsymbol{H}^t = E^t(\boldsymbol{t}), \tag{7}$$

where E^t is BERT type embedding, and H^t is the embedded type sequence.

The dimensions of the elements in H^t are identical to those of the character hidden state vector H^1. The character-type attention module can serve to embed category vectors and output a character-type hybrid vector that blends administrative division hierarchy category information with character hidden state vectors:

$$q_i^t = W_3(ReLU(W_4 h_i + W_t h_i^t + b_4)) + b_3, \tag{8a}$$

$$q^t = \Sigma_{i=1}^n q_i^t \times h_i^t. \tag{8b}$$

The character-type attention operation logic resembles that of the gazetteer-position attention in the gazetteer adapter, with the output q^t representing the weighted character-type representation. This representation integrates hierarchical category features into the administrative division elements in the addresses, where elements of different hierarchical characteristics influence the model's judgment of element categories when parsing addresses.

3.3 GTBERT

GTBERT represents a slightly modified BERT with the gazetteer and type adapters between its first and second layers. A dynamic weighting method combines the outputs of the two adapters into a unified vector. Compared to the BERT encoder, GTBERT only requires updating parameters of the gazetteer and type adapters, and the dynamic weighting module during fine-tuning, with the pre-trained BERT weights preserved. This design constitutes a fully decoupled and parameter-efficient approach. Its effective utilization of external gazetteer and administrative hierarchy knowledge enables the output character representations to encapsulate gazetteer features and administrative hierarchy characteristics. With the character embedding operations and first-layer GTBERT encoder detailed above, we now elaborate on the dynamic weighting mechanism for combining the outputs of the gazetteer and type adapters. Specifically, the first-layer encoder of GTBERT first encodes the address sequence characters to produce hidden states H^1. Then, these hidden states are duplicated and simultaneously fed into the gazetteer and type adapters, generating corresponding hybrid vectors q^g and q^t, as expressed in Equation (9):

$$q^g = \mathrm{GA}(h^1, h^g, R), \tag{9a}$$

$$q^t = \mathrm{TA}(h^1, h^t), \tag{9b}$$

where the hybrid vector and character vector H^1 are transformed into the final representation $\tilde{H}$ via self-attention-based dynamic weighting. This module employs the character-gazetteer hybrid vector q^g and character-type hybrid vector q^t in concatenation as the key and value components. These are then used to compute attention weights with the character vector H^1 before weighted fusion and residual connection with H^1. The process concludes with dropout and layer normalization operations to output $\tilde{H}$. The dynamic weighting is expressed as Equation (10):

$$Q = H^1 W_q, \tag{10a}$$

$$K = \mathrm{CONCAT}(q^g W_{kg}, q^t W_{kt}), \tag{10b}$$

$$V = \mathrm{CONCAT}(q^g, q^t), \tag{10c}$$

$$\tilde{H} = \mathrm{LN}(\mathrm{DO}(H^1 + \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V)), \tag{10d}$$

where W_q, W_{kg}, and W_{kt} are the attention weight matrices, and d_k represents the dimensionality of the key vectors. The attention-weighted output representation $\tilde{H}$ is fed back into the remaining BERT layers for subsequent computations. The computational process for each remaining BERT encoder layer is formulated as Equation (11):

$$M = \mathrm{LN}(H^{l-1} + \mathrm{MultiHeadAttn}(H^{l-1})), \tag{11a}$$

$$H^l = \mathrm{LN}(M + \mathrm{FFN}(M)), \tag{11b}$$

where H^l is the l-th layer output of the BERT encoder, MultiHeadAttn denotes the multi-head attention mechanism, and FFN represents the feed-forward neural network. The final layer output of the BERT encoder is decoded by the efficient global pointer decoder into the address parsing results. In Appendix B, the GTBERT-Pointer algorithm is further introduced to provide a more intuitive demonstration of the inference process, along with a discussion on the computational costs required by each module.

3.4 Efficient global pointer

An efficient global pointer decoder layer is designed to derive the boundary and category dependencies of the address elements from the GTBERT encoder while preserving compatibility with Chinese address parsing tasks involving numerous categories. As a globally normalized pointer network, the efficient global pointer simultaneously determines entity boundaries from a global perspective while determining the category through multi-label classification. Unlike sequence labeling, this approach particularly suits address parsing tasks requiring extensive categories. The efficient global pointer enhances parameter utilization by unifying the extraction process scoring matrices across all categories into a single matrix, while incorporating word segmentation information during loss calculation to ensure training-prediction consistency.

The efficient global pointer differs from other decoders in the scoring and loss functions. The implementation of the scoring function is presented first. Since all entities are continuous, the maximum number of candidate toponym elements in an address sequence with a length of n is $n(n+1)/2$. Since address parsing is essentially filtering genuine entities from this candidate set, the NER task involving sentences with k entities and m entity types can be modeled as selecting k entities from $n(n+1)/2$ candidates while performing m-class classification for each entity. Specifically, with the hidden state sequence $H^e = \{h_1^e, h_2^e, \ldots, h_n^e\}$ from the BERT encoder for address representation, the hidden states first undergo two linear transformations to produce the query and key vector sequences: $q_\alpha = \{q_{1,\alpha}, q_{2,\alpha}, \ldots, q_{n,\alpha}\}$ and $k_\alpha = \{k_{1,\alpha}, k_{2,\alpha}, \ldots, k_{n,\alpha}\}$. Finding the inner product of these vectors yields the scoring function for entity type α with start and end positions x and y in the address element:

$$s_\alpha(x, y) = q_x^T k_y + w_\alpha^T [q_x; k_x; q_y; k_y], \tag{12}$$

where w_α denotes the identifier for entity type α, $q_x^T k_y$ represents the scoring matrix, and $[q_x; k_x; q_y; k_y]$ constitutes the span representation of the hidden states from x to y.

Considering the high sensitivity of the scoring matrix to entity lengths and their positions within sentences, additional positional information becomes critically important for the scoring function. Thus, the efficient global pointer employs Rotary Position Embedding (RoPE) to encode the relative positional information of address element α within the sequence. The matrix transformation $R_x^T R_y = R_{y-x}$ effectively encodes the relative positional relationships between consecutive spans x and y. Incorporating this transformation matrix into the scoring matrix derives an enhanced scoring function, as represented in Equation (13):

$$s_\alpha(x, y) = q_x^T R_{y-x} k_y + w_\alpha^T [q_x; k_x; q_y; k_y], \tag{13}$$

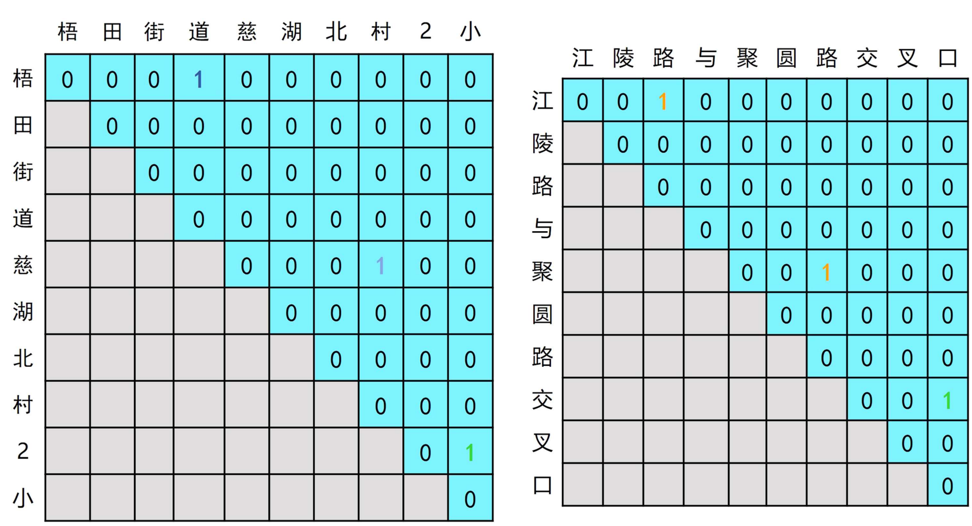
With the scoring function of the efficient global pointer established, the loss function design is presented next. The high total number of categories, substantial quantity of target categories, and larger number of high-level address elements than low-level ones often render Chinese address parsing tasks into imbalanced classification problems. The entity prediction function from the global pointer formulates address parsing as a multi-label classification task. Conventional multi-label classification employs binary classifiers with softmax and cross-entropy loss, prone to computational complexity and suboptimal performance. To address these limitations, a multi-label loss function combining softmax with cross-entropy is adopted for the efficient global pointer. This loss function extends probability calculation and cross-entropy loss to multi-label scenarios. As a result, the class imbalance issue is mitigated via pairwise comparisons between scores of the target and non-target categories. The mathematical formulation of this loss function is expressed as Equation (14):

$$\mathscr{L}_{AP} = \log\left(1 + \sum_{(x,y)\in P_\alpha} e^{-s_\alpha(x,y)}\right) + \log\left(1 + \sum_{(x,y)\in Q_\alpha} e^{s_\alpha(x,y)}\right), \tag{14}$$

where P_α denotes the set of all start-end pairs of address element type α within the x-y range, while Q_α is the set of non-address element type α pairs. Any decoding result with $s_\alpha(x, y) > 0$ identifies the segment into entity α. Simple examples of entity type prediction are depicted in Figure 4. To keep the training and prediction processes consistent, the efficient global pointer entity score loss calculation is augmented with an additional loss term for word segmentation scores, implemented via the fine-grained Chinese word segmentation module in HanLP. The total loss combines the multi-label classification loss with the binary classification loss for word segmentation, which is formulated in Equation (15):

$$\mathscr{L} = \mathscr{L}_{AP} + \lambda \mathscr{L}_{TS}, \tag{15}$$

where $\mathscr{L}_{AP}$ is the loss for Chinese address parsing, $\mathscr{L}_{TS}$ represents the loss based on external token segmentation, and λ is the weight of the segmentation loss. Empirical testing shows that setting λ to 0.01 yields the most consistent prediction results.



case1: No.2 Primary School, Cihu North Village, Wutian Street

case2: Intersection of Jiangling Road and Juyuan Road

Figure 4. The illustration of entity type prediction, where the row index indicates the start of an address entity, while the column index indicates its end.

4 Experiments

This section presents the comparative experiments on two Chinese address parsing task datasets to evaluate the performance improvement of the proposed GTBERT-Pointer against existing address parsing baselines. Ablation studies to assess the significance of contributions and the computational overhead from different modules are also detailed.

4.1 Experimental setup

4.1.1 Evaluation metrics

The performance in Chinese address parsing tasks was evaluated by comparing the output results with the annotations based on three standard evaluation metrics: precision (P), recall (R), and F1-score (F1).

$$P = \frac{\sum_{n \in N} |C_n \cap G_n|}{\sum_{n \in N} |G_n|}, \tag{16a}$$

$$R = \frac{\sum_{n \in N} |C_n \cap G_n|}{\sum_{n \in N} |C_n|}, \tag{16b}$$

$$F1 = 2 \times \frac{P \times R}{P + R}, \tag{16c}$$

where N is the total number of test instances, with one address input denoted as n, which contains x elements corresponding to T labels: $C_n = c_1, c_2, c_3, \ldots, c_x$. The model's output contains y elements corresponding to T labels: $G_n = g_1, g_2, g_3, \ldots, g_y$.

4.1.2 Datasets

Two open-source Chinese address parsing datasets from different periods were selected in this study. The CADD[2] dataset originates from the study entitled 'Neural Chinese Address Parsing', while the GeoETA[1] dataset serves in one of the evaluation tasks in the Geographic Language Understanding Evaluation Benchmark (GeoGLUE). Both the original datasets were of the sequence labeling type. As the efficient global pointer simultaneously predicted both the entity span and the entity category, each entity must have corresponding information about its start and end positions in the text, entity type, and literal content, a span-based named entity annotation methodology. This approach mandated dataset cleaning and transformation, while our data preprocessing script iteratively processed each instance in the training, testing, and validation sets. By analyzing the positional distribution characteristics of the entities indicated by tags across spatiotemporal coordinates, their initiation and termination positions were systematically determined. Accordingly, the essence type of each entity was resolved through a comprehensive analysis of tag attributes throughout multilayered processing modules. Ultimately, the textual data were restructured into the standardized JSON format dataset using biologically inspired discontinuous concatenation mechanisms combined with machine-understandable specification fulfillment techniques. Table 1 presents the statistical information and post-processed span samples of both datasets.

Table 1. Statistical information of the CADD and GeoETA datasets.

Dataset	Type	Train	Dev	Test	Examples	Labels
CADD	Sentence	8.96k	2.99k	2.99k	淳化街道陶苑111栋837(Room 837, Building 111, Taoyuan, Chunhua Street)	('town',0,3,'淳化街道'),('poi',4,5,'陶苑'), ('houseno',6,9,'111栋'),('roomno',10,12'837')
	Token	150.02k	50.16k	49.56k		
GeoETA	Sentence	7.97k	1.970k	0.89k	广东省广州市海珠区上冲村(Shangchong Village, Haizhu District, Guangzhou City, Guangdong Province)	('prov', 0, 2, '广东省'), ('city', 3, 5, '广州市'), ('district', 6, 8, '海珠区'), ('community', 9, 11, '上冲村')
	Token	137.03k	33.18k	14.93k		

- CADD: This dataset comprises publicly accessible addresses collected by crawlers from Chinese social media platforms and online map API services. The training, validation, and test sets follow a 6:2:2 ratio. Following the BIO annotation schema, the dataset categorizes Chinese address components into 21 chunk labels: country, province, city, district, town, village, road, local area, house number, building number, etc.
- GeoETA: Proposed as one of the six subtasks in GeoGLUE for evaluating geolinguistic comprehension capabilities, this dataset primarily comprises addresses for logistics and food delivery. It covers administrative divisions (province, city, district, and town), road networks (road and road number), geographic details (POI, house number, room number, and distance), and non-geographic elements (redundant, others), utilizing the BIOSE annotation with 17 chunk labels. Since GeoGLUE does not include a test set, 10% of the training data is randomly selected to construct the testing set.

4.1.3 Implementation details

The GTBERT-Pointer was implemented as a standard 12-layer BERT model using PyTorch v2.2 and the transformer framework. The training was performed on a platform with a single NVIDIA Tesla V100 GPU featuring 16 GB of GRAM (peak GRAM usage 8 to 16 GB). To control variables, the pre-trained model weights in our proposed model were initialized with the bert-base-Chinese weights provided by Google through Hugging Face[3], keeping consistent all BERT weights in control models. The hyperparameters affecting performance during implementation were categorized into four components: training process parameters, BERT model parameters, BERT adapter parameters, and dataset-specific parameters, with identical hyperparameter settings maintained across both datasets. The training protocol comprised 10 epochs with an initial learning rate of 1×10^{-5}, while the BERT adapter and efficient global pointer components employed a learning rate of 1×10^{-3}. The maximum sequence length and training batch size were set to 128 and 24, respectively. The transformer encoder layers and hidden size dimensions of the BERT model remained consistent with the original pre-trained configuration. For the implementation of the BERT adapter, the maximum number of gazetteer entries per sentence embedding was set to 3, with gazetteer word vectors dimensioned at 200, and the adapter was inserted behind BERT's first layer. Detailed hyperparameter configurations are presented in Table 2.

4.1.4 Baselines

The following state-of-the-art approaches were selected to thoroughly evaluate GTBERT-Pointer, which leveraged neural networks for Chinese address parsing or sequence labeling methods enhanced with lexicons or Global Pointer.

- APLT (Li et al. 2019): A Chinese address parsing model based on a variable tree structure and chart-based neural networks.
- BERT-CRF (Zhang et al. 2020): Featuring a pre-trained BERT model combined with an additional CRF to label address elements in the address sequence.
- RoBERTa-BiLSTM-CRF (Zhang, Luo, and Wu 2023): An improvement over BERT-CRF by substituting the BERT weights with RoBERTa weights and adding a BiLSTM layer between the BERT and CRF layers to refine the representations.

Table 2. Major Hyperparameters.

Parameter	Value	Parameter	Value
Layer of BERT encoder	12	Hidden size	768
Word embed dim	200	Epoch number	10
Max word num	3	Batch size	24
Add layer	1	Learning rate	1e−5
Weight decay rate	0.01	Other learning rate	1e−3
Epsilon rate	1e−8	Max seq length	128
Warmup proportion	0.1		

- MGEO (Ding et al. 2023): A pre-trained multi-task geographic language model with a geographic encoder and multi-modal interaction module. Fine-tuned versions of this model enable tasks such as geographic composition analysis, geographic textual similarity reranking, geographic entity alignment, and address entity alignment.
- Dynamic FSM (Zhang et al. 2023b): A dynamic FSM that adapts its states based on address element types for non-standard Chinese address parsing, integrating TCNN, BiLSTM, and IDCNN for extraction.
- CapICL (Ling et al. 2023): With an REB-KNN algorithm for sample selection based on regular expression matching and BERT semantic similarity, which achieves low-resource Chinese address parsing by prompting a GPT−3 model.
- Global Pointer (Su et al. 2022): The original Global Pointer method for NER, which unifies the recognition of nested and flat entities into one task and introduces RoPE to include relative position information in sequences explicitly.
- FLAT (Li et al. 2020): With a lossless method for incorporating lexicon information into NER tasks, which cleverly uses relative positions to merge lattice structures by modifying the attention mechanism.
- SoftLexicon (Ma et al. 2020): It incorporates lexicon information at the character representation level without affecting the model structure, leveraging the powerful pre-trained representations of BERT to enhance performance.
- LEBERT (Liu et al. 2021a): The first method to apply a lexicon adapter to NER tasks, leveraging it to integrate external lexicon knowledge with the hidden states from the lower layers of BERT to enhance the lexicon representation of character sequences.

4.2 Results and analysis

This section further verifies the improvements of GTBERT-Pointer on Chinese address parsing datasets through comparative experiments with other baselines and ablation studies, and the factors driving the improvements are preliminarily analyzed.

4.2.1 Comparative experiment

A critical comparative experiment was conducted with existing baseline models and GTBERT-Pointer under identical settings and test datasets (CADD and GeoETA). The specific results are presented in Table 3. Overall, GTBERT-Pointer outperformed previous state-of-the-art models across two distinct Chinese address parsing tasks, demonstrating significant improvements. These findings further highlight the effectiveness of external gazetteers and administrative division knowledge, as well as the efficient global pointer mechanism, in enhancing address parsing.

Baselines lacking a pre-trained BERT model (APLT, FLAT, and SoftLexicon) performed the poorest, reflecting the significant enhancement from BERT's text representations for address parsing. Merely employing CRF to decode BERT representations through sequence labeling achieved

Table 3. Main results (precision, recall, and F1 score) with the proposed GTBERT-Pointer and different baselines on the CADD and GeoETA datasets.

	CADD			GeoETA		
Models	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
APLT	90.65	89.21	89.93	90.39	89.27	89.83
FLAT	89.30	89.61	89.46	89.66	89.51	89.58
SoftLexicon	89.63	88.98	89.31	91.00	91.43	91.21
BERT-CRF	89.59	90.33	89.96	91.01	91.73	91.37
SoftLexicon-BERT	90.17	89.97	90.07	91.53	91.89	91.71
RoBERTa-BiLSTM-CRF	89.93	90.33	90.13	91.10	92.57	91.83
MGeo	89.21	90.88	90.04	90.33	92.73	91.51
LEBERT	89.83	90.27	90.05	90.97	92.45	91.70
Global Pointer	90.36	90.74	90.64	91.61	90.38	91.98
Dynamic FSM	90.68	90.65	90.66	92.28	91.79	92.03
CapICL	93.39	89.74	91.53	–	–	–
GTBERT-Pointer	92.25	93.29	92.77	94.65	91.83	93.22

competitive address parsing performance: BERT-CRF attained an F1 score of 89.96% (CADD) and 91.37% (GeoETA). Simply replacing SoftLexicon's static character representations with BERT embeddings yielded a 0.6% improvement on average. With a more powerful encoder, RoBERTa, and a selective forgetting intermediate layer, RoBERTa-BiLSTM-CRF surpassed BERT-CRF by 0.09% on CADD and 0.46% on GeoETA. As a geographic language model pre-trained with POI and coordinate data to enhance geo-related tasks by leveraging POI address semantics and spatial semantics, MGeo achieved notably better performance (90.04% on CADD and 91.51% on GeoETA). LEBERT introduced a lexicon adapter with BERT weights to improve performance in Chinese NER tasks, demonstrating competitive results on CADD (90.05% F1) and GeoETA (91.70% F1). However, LEBERT exhibited limited performance improvement in address parsing due to its corpus mismatch for address-related tasks and lack of relative positional information for candidate words during feature fusion. Global Pointer emphasized global semantic features when processing complete address strings, achieving a 90.64% F1 on CADD (0.68 percentage points higher than BERT-CRF) and a 91.98% F1 on GeoETA (0.61 percentage points higher than BERT-CRF). Thus, it demonstrated an inherent suitability for Chinese NER tasks with numerous entity categories and large entity quantities. Dynamic FSM integrates address domain knowledge in the form of dynamic finite states upon the RoBERTa-BiLSTM-CRF foundation, achieving F1 scores of 90.66% and 92.03% on the CADD and GeoETA datasets, respectively. On average, its performance represented an improvement of 0.3 percentage points over RoBERTa-BiLSTM-CRF. Conversely, CapICL employed BERT to match similar addresses for constructing sample prompts that guide GPT models in address parsing, attaining a 91.53% F1 score on CADD. However, CapICL still relied on BERT's semantic matching capability, and the GPT−3 model's weights exceed those of BERT by over a thousand times. As a result, substantial computational resources were consumed during address parsing inference.

GTBERT-Pointer integrates external gazetteer features and administrative division features through its adapters and employs Global Pointer as a decoder more suitable for address parsing, effectively combining their advantages. Thus, it achieved significant improvements across all datasets. By improving the decoder and deriving an efficient global pointer that unifies the scoring matrices during classification and incorporates word segmentation information into loss calculation, GTBERT-Pointer outperformed Global Pointer by 2.13 percentage points (92.77% vs. 90.64%) on the CADD dataset and 1.24 percentage points (93.22% vs. 91.98%) on the GeoETA dataset. These results demonstrate that the gazetteer and type adapters can enhance model performance to varying degrees while highlighting the advancement of the efficient global pointer. Compared with LEBERT, GTBERT-Pointer achieved improvements of 2.72 and 1.52 percentage points on the two datasets, respectively, while demonstrating varying degrees of enhancement over MGeo (2.73 and 1.71 percentage points). The most significant improvements of GTBERT-Pointer over the above models were attributed to the efficient global pointer, followed by the replacement of LEBERT's lexicon adapter with the gazetteer and type adapters. The gazetteer and administrative division data proved more conducive to address parsing than the POI data in MGeo. Furthermore, GTBERT-Pointer outperformed the second-best model, CapICL, by 1.24 percentage points (91.53 vs. 92.77) on the CADD dataset, indicating that the increased computational resources of GPT−3 relative to the BERT models do not translate into better address parsing performance. In contrast, LLMs are more suitable for multi-tasking and complex reasoning. GTBERT-Pointer also outperformed Dynamic FSM by 2.11 and 1.19 percentage points on the two datasets, confirming that the gazetteer and type adapters can more effectively incorporate address domain knowledge into BERT than dynamic finite state methods.

4.3 Ablation study

To elucidate how its modules influence GTBERT-Pointer, this section presents ablation studies by removing the critical gazetteer adapter, type adapter, and efficient global pointer in turn. The experiments were based on identical hyperparameters and datasets. Random seeds were modified for five independent trials, and the results were averaged. As shown in Table 4, the consistent trends of

Table 4. Ablation studies to evaluate the proposed model without the adapters or the efficient global pointer. GA and TA stand for the gazetteer and type adapters, and EGP stands for the efficient global pointer. 'w/o' is the abbreviation for 'without,' indicating the absence of a module.

Settings	GeoETA F1(%)	CADD F1(%)	CADD Time(s)/epoch	Params(M)	FLOPs(G)
Baseline(GTBERT-Pointer)	93.22	92.77	80.10	110.66	120.13
- w/o GA	92.20	92.18	76.19	103.00	119.29
- w/o TA	93.07	92.42	78.37	110.07	120.04
- w/o Adapters	92.16	90.79	75.07	102.41	119.30
- w/o EGP	92.73	91.71	80.08	110.56	119.88
- w/o Adapters and EGP	91.37	89.96	76.38	102.31	119.05

the performance across both datasets confirm the systematic improvements from individual modules. First, removing the gazetteer adapter reduced F1 scores by 1.12 percentage points on CADD and 0.59 percentage points on GeoETA, whereas excluding the type adapter resulted in smaller declines of 0.15 and 0.35 percentage points, respectively. Therefore, the gazetteer adapter is more important for geographic element extraction. This discrepancy may have arisen because the gazetteer corpus already contained vector representations for most administrative divisions, while the additional hierarchical vectors for administrative levels only marginally affected the geographic element extraction process without significantly influencing category determination. Removing the entire adapter module caused the most significant degradation (1.06 percentage points on CADD and 1.98 percentage points on GeoETA), underscoring its importance. Meanwhile, the combined effect of the gazetteer and type adapters exceeded the contribution of either individual component. Notably, the results also demonstrated the significance of the efficient global pointer. When replaced with the commonly used CRF decoding method for sequence labeling, the model without an efficient global pointer exhibited reductions of 0.49 percentage points on CADD and 1.06 percentage points on GeoETA, confirming its indispensability in the proposed framework.

While various modules enhance the overall performance (a crucial indicator), they inherently increase computational costs while improving prediction accuracy. Thus, the model complexity and computational costs were compared across different configurations to analyze module impacts. Quantified metrics adopted included total model parameters (Params), floating point operations per second (FLOPs), and average training time per epoch on the CADD dataset.

As shown in Table 4, FLOPs, an indicator of the computational load, vary with model parameters under identical inputs and platforms. Model parameters correlated positively with training duration. Removing the type adapter slightly reduced parameters, whereas eliminating the gazetteer adapter yielded larger reductions. Replacing the efficient global pointer with a CRF layer increased the parameters due to CRF's additional linear mapping layer. The type adapter introduced minimal overhead through character-type attention, while the gazetteer adapter's additional spatial attention and linear layers for dimensional alignment substantially increased the parameters. The efficient global pointer was lighter than CRF and contributed to faster training. Most parameters originated from the 12-layer BERT backbone, and GTBERT-Pointer increased the average epoch time by less than 10 seconds compared to BERT-CRF.

5 Discussion

5.1 Mechanism analysis

Based on the results of ablation studies in Table 4, the mechanisms by which the gazetteer and type adapters enhance model generalization were analyzed, particularly in terms of mitigating overfitting and handling OOV issues. Removing the gazetteer adapter (w/o GA) led to larger F1 drops (1.02 percentage points on GeoETA, 0.59 percentage points on CADD) than removing the type adapter (w/o TA; 0.15 and 0.35 percentage points), underscoring its greater contribution. This disparity was possibly because the gazetteer adapter, through its gazetteer-position attention, infused external gazetteer and geographic knowledge to mitigate OOV issues from imbalanced distributions and informal expressions. Dynamically integrating pre-computed embeddings from the gazetteer corpus allowed it to reduce BERT's reliance on learned patterns from limited training data. As a result, overfitting was curbed, and the robustness to regional variations was improved more effectively than baselines like BERT-CRF, which lacked such domain-specific infusion. In contrast, the type adapter primarily captured hierarchical relationships via character-type attention and tree-structured retrieval of administrative divisions, which facilitated semantic disambiguation but contributed less to OOV handling. Its smaller impact revealed by the ablation studies suggests that it complements rather than drives model generalization, especially for datasets with prevalent unstructured formats. Overall, these adapters balanced neural network flexibility with rule-based constraints, outperforming traditional address parsing approaches through their decoupled, parameter-efficient knowledge enhancement.

5.2 Hyperparameter analysis

This section discusses the effects of the new hyperparameters introduced by the BERT adapters on the performance relative to the original model, and the methods to identify their optimal values are explored. While maintaining identical settings for all other hyperparameters, we systematically varied either the maximum candidate sequence length or the adapter insertion positions between BERT layers to observe the F1-score variations on the CADD dataset. The results could help determine the optimal configurations for the gazetteer and type adapters. Table 5 demonstrates that setting the maximum candidate number to 3 yields the highest F1-score. This optimal balance emerged because a single candidate provided no alternative segmentation options (risking erroneous segmentation), whereas five candidates introduced excessive noise with detrimental effects outweighing the benefits of enriched candidate features. Experimental findings indicated that inserting a single adapter layer at the deepest BERT layer achieved optimal performance. Placing the adapter at shallower layers or inserting multiple adapters degraded model effectiveness. We hypothesize that introducing adapters at shallower BERT layers or including multiple adapters may cause excessive reliance on external gazetteer and administrative division features, granting these explicit feature vectors dominance over the latent representations learned by the model.

In addition to ablation studies, we reassessed the performance of GTBERT-Pointer after replacing its efficient global pointer with the original Global Pointer and CRF. Thus, we can specify the manifestations of 'efficient' in the efficient global pointer. As shown in Table 6, the efficient global pointer significantly outperforms both the Global Pointer and CRF on the two datasets. However,

Table 5. GTBERT-Pointer performance with varying maximum matched words and adapters inserted at different layers.

Number of words	Layer of insertion	F1(%)
1	1	92.65
3	1	92.77
5	1	92.69
3	6	85.76
3	12	46.42
3	2,4,6	71.70
3	2,6,12	15.92
3	8,10,12	6.74
3	2,4,6,8,10,12	0.99

Table 6. Address parsing performance, model complexity, and computational costs of GTBERT-Pointer with the efficient global pointer, Global Pointer, or CRF.

	GeoETA	CADD			
Settings	F1(%)	F1(%)	Time(s)/epoch	Params(M)	FLOPs(G)
Efficient global pointer	93.22	92.77	80.10	110.66	120.13
Global pointer	92.89	91.59	80.10	114.39	125.05
CRF	92.16	90.79	75.07	102.41	119.30

Global Pointer surpassed CRF by only 0.06 percentage points in F1-score on the GeoETA dataset while slightly underperforming CRF on the CADD dataset. Although Global Pointer had an identical training duration to the efficient global pointer, its Params was 1.03 times that of the efficient global pointer, and its FLOPs was 1.04 times that of the efficient global pointer. In summary, the 'efficient' aspect of the efficient global pointer was demonstrated through its consistent enhancement of address parsing performance and alignment with the total parameters of a lightweight model.

5.3 Limitations and future work

Despite its advances in domain knowledge integration, adapter decoupling, parameter efficiency, and multi-label adaptability, GTBERT-Pointer still showed notable limitations. Its reliance on static gazetteer corpora makes it vulnerable to outdated data. Although regular updates can mitigate performance degradation due to evolving address patterns, they incur substantial maintenance costs. The adapters struggle with highly ambiguous inputs, such as addresses lacking administrative levels or using free-form intersections, which deviate from hierarchical structures. Performance on dialectal addresses remains unexplored and may intensify OOV challenges. While the efficient global pointer mitigates sequence length issues, scaling to ultra-long addresses still raises computational demands. Finally, although modest (4 to 5 M parameters, <10 seconds per epoch; Table 4), adapter overhead may constrain deployment in resource-limited scenarios.

LLMs represent a promising avenue to enhance GTBERT-Pointer. For instance, LLMs can generate contextual features to augment the adapters, providing richer semantic priors for disambiguating informal expressions or OOV terms, as demonstrated by recent frameworks for address standardization. For highly ambiguous cases, where the pointer decoder yields low-confidence spans, employing LLMs as a backoff mechanism can refine parses through chain-of-thought prompting or retrieval-augmented generation. This hybrid approach would further the proposed model's parameter-efficient design, potentially improving its generalization without retraining.

Thus, future work may incorporate dynamic updating mechanisms for the gazetteer and type adapters, such as integrating APIs for real-time administrative data retrieval. Enhancing ambiguity handling through hybrid rule-based preprocessing or multi-modal integration with map data (e.g. fusing geospatial embeddings from external sources) may further improve robustness. Additionally, integrating dialect alias databases may solve issues induced by regional linguistic variations, and geographic coordinate-embedded representations may enable spatial semantic understanding, potentially enabling multi-task approaches encompassing address matching and geocoding. For low-resource scenarios, few-shot LLM prompting may facilitate adaptation to new domains with minimal annotated data, extending the model's applicability to underrepresented regions or languages.

6 Conclusion

This study addresses Chinese address parsing challenges stemming from rapid data updates, flexible address patterns, and significant regional variations with GTBERT-Pointer, a BERT encoder enhanced with gazetteer and administrative hierarchy knowledge through adapters and an efficient global pointer for decoding. Experiments demonstrate that GTBERT-Pointer significantly outperforms the baselines. Thus, the domain-specific knowledge introduced via the adapters in a flexible and parameter-efficient manner enhances address processing capabilities, whereas the efficient global pointer approach outperforms other sequence-labeling decoding methods for multi-category address parsing. Results of ablation studies prove our effective architectural choices with the designed gazetteer-position attention (integrating gazetteer features into character embeddings) and character-type attention (incorporating administrative hierarchy features into character hidden states). The architecture effectively constrains model behavior while enhancing interpretability for domain-specific tasks. The findings also indicate that sharing scoring matrices across all categories during prediction and incorporating pre-segmentation results into the multi-label classification loss significantly improve Global Pointer's efficiency. As a more practical solution, GTBERT-Pointer could improve government census registration, traffic management, and logistics by advancing the structured processing of address texts.

Endnotes

1. CADD: https://github.com/leodotnet/neural-chinese-address-parsing, GeoETA: https://modelscope.cn/datasets/damo/GeoGLUE/summary
2. Geonames wikipedia database: https://www.geonames.org/export/wikipediawebservice.html
3. China's 2023 administrative divisions document: https://www.stats.gov.cn/sj/tjbz/tjyqhdmhcxhfdm/2023/index.html
4. BERT pre-trained model weights: https://huggingface.co/google-bert/bert-basechinese

Notes on contributors

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Disclosure statement

The authors declare that there is no conflict of interest.

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Data and codes availability statement

The data presented in this paper are available upon request from the corresponding author.

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Appendices

Appendix A Gazetteer embedding corpus data collection and cleansing details

We leveraged Wikipedia web services[4] to acquire geographic names and descriptions, which involved accessing georeferenced Wikipedia articles in 240 languages. Each record contained fields like title, summary, geographic coordinates, and countryCode. Data collection was performed using the Wikipedia Articles with the Bounding Box API, which employed bounding box coordinates based on the level−3 S2 spatial index grid. We systematically traversed all bounding boxes covering China's territory to obtain toponym-related data. Initial country code filtering removed geographic entries for peripheral nations, followed by data cleansing procedures like deduplication, removing entries lacking descriptions, stripping HTML tags, and converting Traditional Chinese to Simplified Chinese for standardized input.

To build the embedding training corpus, we extracted titles as core terms and summaries as contextual information. The Jieba text segmentation tool served to process contextual text, with each entry formatted as 'geographic name + summary' to enhance domain-specific representation. Unlike traditional Word2Vec, FastText with Skip-gram training strategy decomposes words into subwords (e.g. `北京市' into `北', `京', and `北京市'), thus improving the handling of rare toponyms and neologisms. After fifty epochs of unsupervised training, a gazetteer corpus G was derived, containing 533,824 geographic entries with corresponding embeddings.

APPENDIX B Algorithm of the proposed GTBERT Pointer Model

Algorithm 1. GTBERT_Pointer Model

Require: Input address sequence $\mathbf{a}$, word IDs $\mathbf{w}$, administrative region IDs $\mathbf{t}$, mask $\mathbf{M}$

Ensure: Entity span predictions $\mathbf{Y} \in \mathbb{R}^{C\times n\times n}$

// **BERT**

$\mathbf{H}^{(0)} = \text{BERTEmbedding}(\mathbf{a})$ // *Cost:* $\mathcal{O}(n \cdot d)$

$\mathbf{E}_t = \text{TypeEmbedding}(\mathbf{t})$ // *Cost:* $\mathcal{O}(n \cdot d)$

$\mathbf{E}_w = \text{GazetteerEmbedding}(\mathbf{w})$ // *Cost:* $\mathcal{O}(n \cdot m \cdot d_w)$

for $i = 1$ to 12 **do**

  $\mathbf{H}^{(i)} = \text{BERTLayer}_i(\mathbf{H}^{(i-1)}, \mathbf{M})$

  if $i = 1$ **then**

    // **Type Adapter**

    $\mathbf{T} = \text{TypeAdapter}(\mathbf{E}_t)$ // *Cost:* $\mathcal{O}(n \cdot m \cdot d^2)$

    // **Gazatteer Adapter**

    $\mathbf{H}^{(i)} = \text{GazatteerAdapter}(\mathbf{H}^{(i)}, \mathbf{E}_w, \mathbf{M}_w)$ // *Cost:* $\mathcal{O}(n \cdot d^2)$

    // **Dynamic weighting module**

    $\mathbf{H}^{(i)} = \text{DynamicW}(\mathbf{H}^{(i)}, \mathbf{T})$ // *Cost:* $\mathcal{O}(n \cdot d)$

  end if

end for

// **EfficientGlobalPointer decoding**

$\mathbf{L} = \text{EfficientGlobalPointer}(\mathbf{H}^{(12)}, \mathbf{M})$ // *Cost:* $\mathcal{O}(n^2 d_h + ndC)$

Overall Complexity:

$$\mathcal{O}(\underbrace{12 \times (n^2 d + 4nd^2)}_{\text{BERT}} + \underbrace{n \times m \times d^2}_{\text{Gazatteer}} + \underbrace{n \times d^2}_{\text{Type}} + \underbrace{n \times d}_{\text{DynamicW}} + \underbrace{n^2 d_h + ndC}_{\text{EGP}})$$

return $\mathbf{Y}$